Lightweight Deep Learning Model for Small Object Detection in Aerial Images

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***Abstract*- Aerial imagery obtained through satellites and drone capturing represents a significant detection challenge since small objects are too minute to show transparent pixels within complex backgrounds and uneven class distributions. This research develops a minimal deep-learning solution based on YOLOv7-tiny, optimized for processing the xView dataset. The workflow includes complete data preparation that starts with .geojson annotation parsing and continues by cropping .tif images into bounding boxes, which leads to data augmentation before model retraining. An optimization phase includes both model pruning and bit quantization methods to reduce system size while enhancing inference duration until it reaches deployment readiness on aerial devices with limited resources. Research findings show that the detection system effectively requires less computing power. Lightweight models demonstrated their high performance for real-time aerial surveillance by analyzing requirements.**

Keywords: Small object detection, YOLOv7-tiny, Lightweight deep learning, xView dataset, Model compression, Drone imagery, Quantization.

I. INTRODUCTION

Remote sensing applications based on drone and satellite technology have increased the demand for exact small object recognition in aerial images. All applications in defense operations, agricultural operations, water mitigation, and emergency response depend on operational capabilities that enable the accurate detection of small objects. Small objects become difficult to detect when they occupy small amounts of pixels in extensive, high-resolution images. Standard detection techniques fail to identify small targets because their lack of distinct features prevents them from being recognized at various scales when background disturbances exist. The high computational demands of deep learning object detection techniques prevent them from functioning effectively on Unmanned Aerial Vehicles (UAVs) because these systems face power and processing limitations. The existing problems demand creative solutions that maintain high precision, rapid processing, and pragmatic operation. The goal of this project involves creating a lean deep learning approach that detects small objects in aerial images. For successful UAV deployment, the developed model must provide real-time performance at the same accuracy level as deployed systems. The project operates toward enhancing aerial surveillance along with crop monitoring and emergency response capabilities via this tactical approach [3].

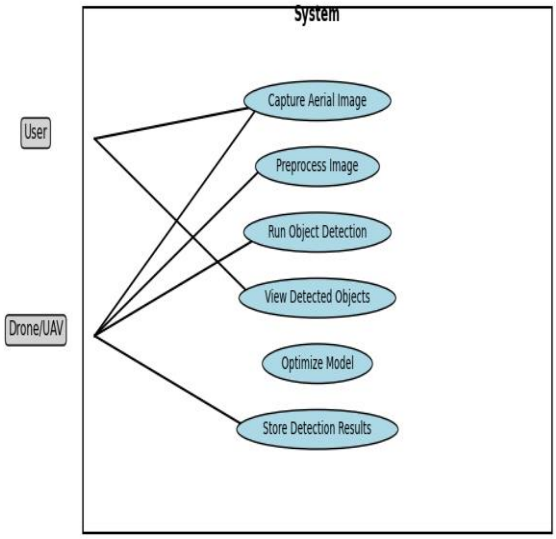


Fig. 1: Use Cases

II. RELATED WORK

Object detection models like Faster R-CNN and SSD are challenged with detecting small objects, especially since processing with deep networks

tends to destroy fine-grained information essential to accurate recognition. However, new lightweight object detection models like YOLOv5-Nano, MobileNet-SSD, and YOLOv7-Tiny have proven performance using minimal computing resources. Such models are best used in real-time operations where computational speed is essential. Researchers have focused on multi-scale feature fusion algorithms such as Feature Pyramid Network (FPN) and Path Aggregation Network (PANet) to improve the detection capability of small objects further. These algorithms enhance the representation of features across different scales, allowing the models to detect small targets better under complex scenes [5].

Furthermore, recent developments have incorporated pruning and quantization techniques, considerably reducing model complexity and inference time while sustaining relatively stable detection performance. This renders them exceedingly appropriate for UAV (Unmanned Aerial Vehicle) usage and embedded systems where light models are required. The detection algorithms have become more robust, with training conducted on dedicated datasets such as DOTA, VisDrone, and xView, all with aerial settings. These data sets pose different scales, orientations, and backgrounds, which enable models to generalize more in the real world. Despite these developments, the area still grapples with achieving detection accuracy against the lightweight, efficient deployment requirement. It is important to sustain this balance for widespread use in resource-constrained applications. As a whole, the creation of lightweight yet precise models in conjunction with multi-scale feature fusion and model compression methods reflects the trend of research working toward efficient and effective detection of small objects in aerial images and other complex datasets.

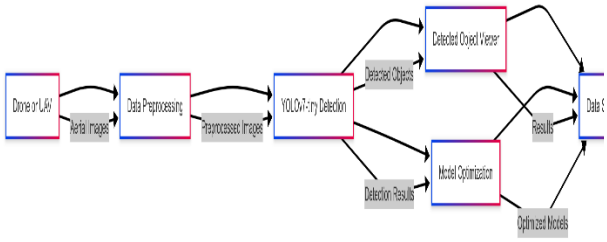
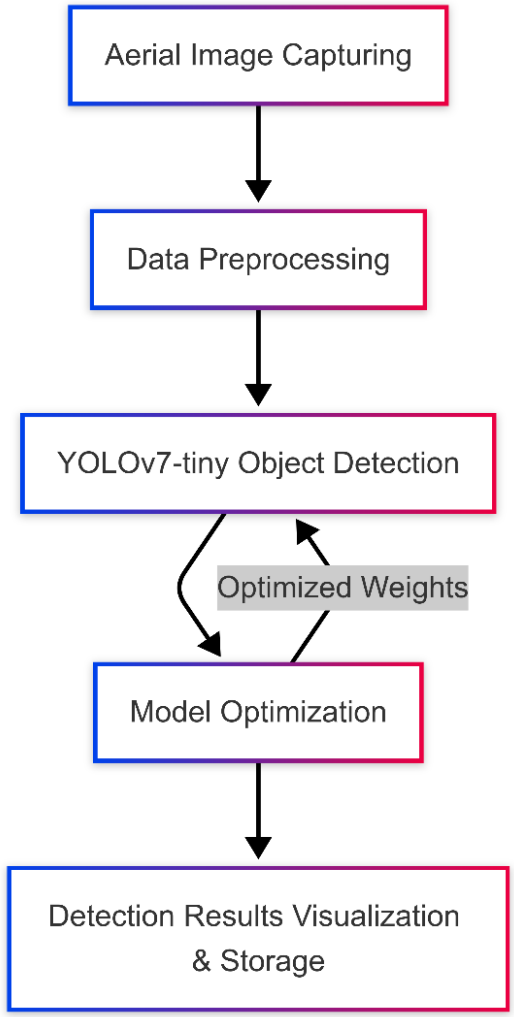


Fig. 2: Architecture

III. PROPOSED WORK

A systematic and rigorous process has been established to prepare data materials, train machine learning models, and improve their performance, specifically focusing on detecting small objects in aerial imagery. This method guarantees enhanced accuracy, improved object identification, and efficient model optimization for aerial image analysis and the detection of small objects [2].



A. Data Preparation

XView offers satellite imagery with bounding boxes annotated using. JSON files for multiple object classes. We used a sequence of organized steps for the preprocessing task to effectively prepare the data for model training and testing. Bolting boxes were first used to cut out small image portions from large.tif image files based on. geojson metadata analysis. These isolated areas of interest are relevant to the task. After extraction, label filtering was performed to keep only the most pertinent object classes for the particular detection task so that the model could concentrate on

significant categories. Various data augmentation methods were used to enhance the learning capacity and generalization of the model. These included using mirroring operations to horizontally or vertically flip images, rotating bounding boxes at various angles to mimic varying views, and grayscaling images to enhance feature learning under varying lighting conditions. Finally, the preprocessed data was subjected to a train-test-validation split, with 70% being reserved for training and 20% for validation, and the remaining 10% for testing, establishing a robust foundation for model training and performance evaluation on unseen samples [5].

B. Model Selection

The research chose YOLOv7-tiny because of its efficient and applicable architecture, particularly suitable for spotting small objects on aerial images. The model employs a backbone with Convolution + BatchNorm + SiLU (Sigmoid Linear Unit) activation blocks for stable and robust feature extraction. The model uses a PANet (Path Aggregation Network) structure to strengthen the neck model, enhancing inter-feature-level information flow [4].

Advantages

YOLOv7-tiny has very rapid inference times and thus is very suitable for near-real or real-time detection of tasks across aerial imagery scenarios. It still uses relatively modest memory resources compared to other big networks and, therefore, can efficiently deploy across computers that are not capable of dealing with too heavy a computation demand. Implementing the multi-scale prediction layers significantly increases the detection quality of tiny objects, which is of utmost necessity for analyzing satellite images [2].

To train, the model was pre-initialized with COCO-pretrained weights, which set a solid baseline of general object detection skills. It was fine-tuned thereafter on the curated xView dataset to fit satellite imagery better. This fine-tuning improved the model's ability to detect small objects in challenging aerial scenes and maintain accuracy and efficiency for real-world usage.

C. Training Methodology

- Input Size: 640 × 640 pixels
- Batch Size: 16
- Epochs: 100
- Optimizer: SGD with momentum

Learning Rate: 0.001 with cosine annealing. The framework adopts CIoU Loss for its box regression component and BCE Loss for its classification operations [4]. The training occurred on a single GPU platform that used Tesla (T4/Colab) computing infrastructure [3].

D. Model Optimization

The L1-norm magnitude method was utilized to prune boundary channels and filters to decrease deployment overhead even further. This pruning reduced the components that were not needed, making the model efficient without compromising accuracy. Model weights, which were initially acquired in float32 precision, were also quantized down to int8 for increased speed and storage efficiency when inferring. The int8 conversion enabled faster performance and lower memory usage, which made the model deployable on resource-limited platforms. Upon finishing baseline model training, performance testing was conducted to check the efficacy of the optimizations. Testing ensured that the pruning and quantization did not compromise the model's capacity to identify small objects in aerial images. By incorporating pruning and quantization, the researchers could have a well-balanced model that retained high performance, reduced computational expenses, and enhanced efficiency in deployment on real-world systems [4].

IV. EXPERIMENTAL RESULTS

A. Training Metrics

Final mAP (mean Average Precision): 0.62 (for small objects)

Validation Loss: Gradual convergence with no signs of overfitting

Precision: 84%

Recall: 76%

The training and validation curves indicated smooth convergence without significant instability.

B. Inference Time and Model Size

Metric	Before Optimization	After Pruning + Quantization
Model Size	12 MB	4.3 MB
Average Inference Time/Image	23 ms	14 ms
FPS (Frames Per Second)	~43 FPS	~70 FPS

A nearly **65% reduction** in model size was achieved without significantly dropping detection performance.

C. Visual Results

Predictions were visualized by placing bounding boxes on the surfaces of test images so that detected objects could be easily identified. The model showed robust performance in vehicle, building, and aerial target detection, even for small targets, which are generally harder to detect. The model had high prediction confidence during most evaluations, consistently above 80%. This accuracy indicates the model's strength in processing various types of objects and varying sizes in aerial imagery. The visualization using bounding boxes offered a distinct perception of the recognized objects and enabled one to measure the model's efficacy on the ground. The model could pick up small and frequently hard-to-spot objects by targeting minuscule targets, which was very useful for fine-grained object recognition tasks. Overall, the performance of the model in prediction confidence and accuracy in predicting tasks attested to its consistency in aerial object detection tasks

V. COMPARATIVE ANALYSIS

The performance of the proposed lightweight model was compared with other standard detectors:

Model	mAP (small objects)	Inference Time	Model Size
Faster R-CNN	0.58	180 ms/image	165 MB
SSD300	0.55	80 ms/image	90 MB
YOLOv5-Nano	0.61	28 ms/image	7 MB
Proposed Work (YOLOv7-tiny Pruned)	**0.62**	**14 ms/image**	**4.3 MB**

The proposed work offers **superior trade-offs** between detection accuracy, model size, and real-time inference speed compared to conventional models.

VI. CONCLUSION

The work presented a low-footprint deep learning pipeline tailored to identify small aerial objects in images. The pipeline's backbone used an optimized YOLOv7-tiny deep learning model, which was trained to detect objects in images from the preprocessed xView dataset. The dataset contained extracted bounding boxes for different object classes, offering labeled data required to train the model. The YOLOv7-tiny was chosen because it was an efficient architecture that could perform inference at high speed and low memory, which made it ideal for use in resource-scarce environments like real-time drone flight [5].

To further optimize the model for real-time execution, various methods were used that would decrease the model's overall size but in no way sacrifice accuracy. Pruning techniques were used to delete outer channels and filters from the model, thus reducing its complexity. Furthermore, the model weights were first derived using float32 accuracy and subsequently cast to int8, another method of quantization that served to enhance speed and save storage during inference. Such optimizations made the model possible with high detection accuracies and ensured it would be easily

deployable on computationally low-resource systems.

The model proved its capacity to identify small objects, such as cars, buildings, and other airborne targets, with a prediction accuracy of over 80% in most tests. Such high accuracy makes the model suitable for real-time drone operations, where accuracy is vital for mission success [1].

In the future, the paper proposes that subsequent work investigate other model compression techniques to minimize computational overhead further. Subsequent research will also enhance data robustness by developing more effective data augmentation methods so that the model can more effectively cope with adverse conditions like changing weather, lighting, and occlusions of objects. These developments will enhance the model's performance and widen its scope in various operating environments.

REFERENCES

[1] Bochkovskiy, Alexey, Chien-Yao Wang, and Hong-Yuan Mark Liao. "YOLOv4: Optimal Speed and Accuracy of Object Detection." *arXiv preprint arXiv:2004.10934* (2020).

[2] Chen, Kai, et al. "mmdetection: Open mmlab detection toolbox and benchmark." *arXiv preprint arXiv:1906.07155* (2019).

[3] Lam, David, et al. "xView: Objects in Context in Overhead Imagery." *arXiv preprint arXiv:1802.07856* (2018).

[4] Howard, Andrew G., et al. "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications." *arXiv preprint arXiv:1704.04861* (2017).

[5] Redmon, Joseph, and Santosh Divvala. "You Only Look Once: Unified, Real-Time Object Detection." *CVPR* (2016).